

Determination of an optimal interpolation technique to represent the spatial distribution of groundwater quality at urban and peri-urban areas of Proddatur, Y.S.R district, Andhra Pradesh, India

Jagadish Kumar. M¹, Sunitha. V², Ramakrishna Reddy. M³

1- Department of Civil Engineering, YSR Engineering College of YVU, Proddatur, AP

2- Department of Geology & Geoinformatics, Yogi Vemana University, Kadapa, AP, India

3- Department of Earth Sciences, Yogi Vemana University, Kadapa, AP, India

jagadishmogaraju@gmail.com

ABSTRACT

The preliminary aim of the present study is to analyze the spatial variability of ground water quality parameters in Proddatur, India. The spatial variability and the GIS maps can be used for management of ground water quality and effective reduction of pollution related hazards can be achieved. Geostatistical analysis is a quintessential commodity to understand the ground water quality parameters and their physico chemical characteristics. The present study was conducted to analyse and assess the groundwater quality from samples collected from 43 bore holes in Proddatur, YSR district of Andhra Pradesh. Physico chemical analysis was conducted on the samples and spatial statistical analysis was performed on the data procured. Geostatistical techniques such as Inverse Distance Weighting (IDW), Radial Basis Function (RBF) and Ordinary Kriging were employed to generate surface maps after selecting the best fit among the three techniques. The key physico chemical parameters such as pH, Total Hardness (TH), Calcium (Ca^{2+}), Magnesium (Mg^{2+}), Chloride (Cl^-), Fluoride (F^-) and Alkalinity were systematically analysed and statistical treatment was conducted on them. A multivariate analysis was also conducted.

Keywords: Ground water quality, Geostatistics, Multivariate analysis, Kriging, IDW, Spline.

1. Introduction

The ground water quality is quintessential for the human growth and welfare. Since the rehabilitation of ground water is difficult if it gets polluted, the effective management of this key resource is inevitable. It is obvious that in current scenario the demand of water for industrial and non industrial needs is increasing in a geometric progression. The spatio-temporal behavior and dynamics of the ground water should be studied in order to maintain an effective management exercise. Stringent necessity to study and evaluate the ground water quality which was polluted due to point and non point sources is essential. In this part of research, the concentration of pollutants existing in unmeasured locations was studied using geostatistical methods. The main objective of the present research was to assess the ground water quality using geostatistical tools. This study was initiated to provide prediction maps and to enlighten the local authorities about the affected areas and susceptible areas which are in the line of deterioration in terms of ground water quality. The Geostatistical methods were employed to model correlation among spatial structures via variogram to determine spatial correlation measurements. Since global statistics incurred pivotal prominence over local statistics, simulation over a stochastic scale is an essential commodity in geospatial analysis. In the sub-surface hydrology studies, errors are to be avoided and enhanced rate of calculation accuracy is expected, so the Geostatistical tools serve as a handy tool to achieve

these tasks. Traditional statistical sampling techniques are devoid of spatial connectivity. In contrary to the classical scenario, Geostatistics takes into account sample values in addition to their location and hence allowing the user to simultaneously study locations and values of the samples collected. The spatial structure is determined by the relating spatial properties such as distance and direction, so as to achieve a meaningful analysis.

2. Study area

The present study area is located southwest of Y.S.R district, India. The area spreads through $78^{\circ}33'E$ to $78^{\circ}55'E$ longitudes and $14^{\circ}44'N$ to $14^{\circ}73'N$ latitudes. This area falls in No-57J/6 toposheet provided by Survey of India. This area has an elevation averaged to 155 meters as per surveys conducted in the recent past. Needless to say this area is flourished with commercial acquaintances and it's considered as second Bombay by the native public. The groundwater resources in this area is vital for urban and peri urban activities prevailing in current situation.

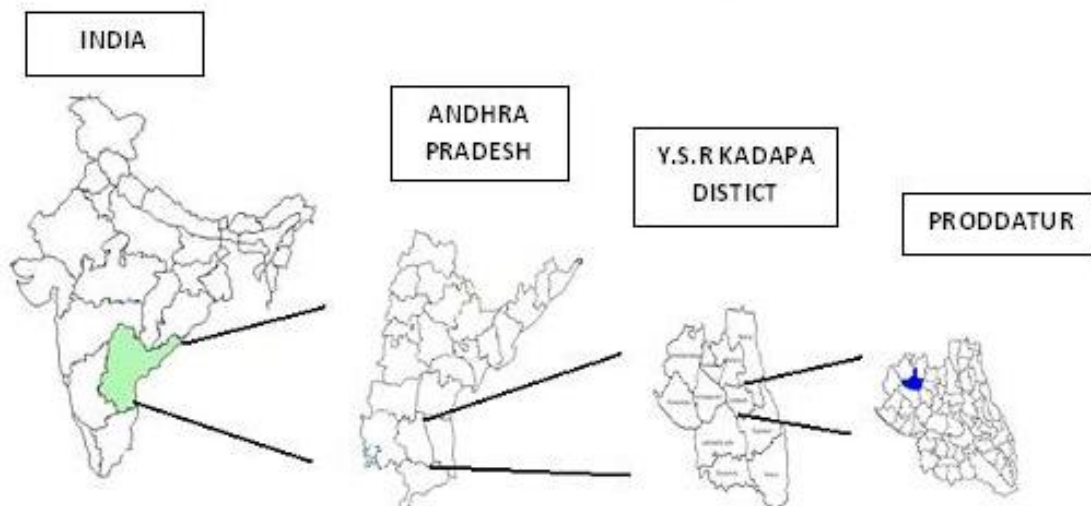


Figure 1: Location map of Study area

3. Methodology

The spatial structure is identified and modeled in the process of geostatistical prediction. The variogram is employed to study homogeneity and spatial continuity of the interested variable. The best fitted variogram decides various stages of this process. The spatial prediction methods include Inverse distance weighting (IDW), Radial basis function (Spline) and Kriging. The distribution and magnitude was evaluated whilst calibrating the data. The present work focused on the hydro geochemical parameters of the local aquifer. Factorial variables are established by employing Principal component analysis and multivariate analysis in order to summarize physico chemical data obtained from the area of interest. Root mean square error was taken into consideration whilst evaluating the performances of the selected models by cross validation mode. The lesser the RMSE values the more the predictions are. The multivariate analysis which includes factor analysis was used in order to establish interrelationship within the data. The major part of the current work is based on two software's namely ARCINFO 9.3 banked with geostatistics module and XLSTAT.

4. Results and discussion

Statistical treatment was employed to produce summary statistics for the ground water quality parameters. The summary statistics were analyzed in non-transformation mode and transformation (log) mode. The summary statistics was represented in table 2 and table 3. The factor analysis was employed to reveal the principal factors which explain the correlations among multiple outcomes. The principal factors are the reduced set of the variables which are generated using a classical method called as principal component analysis. PCA was conducted among 43 bore wells and it's followed by factor analysis. The variance of the loadings within each factor can be achieved by varimax rotation. The correlation matrix was established using the variables. The best fitted interpolation technique among the three for each variable was evaluated based on the RMSE values.

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Table 1: Result of chemical and statistical analysis of physico-chemical parameters of ground water samples collected from the study area.

VILLAGES	PH	TH	Ca	ALKALINITY	Cl	F	Mg
Chowdur	7.90	1692.00	520.00	310.00	2420.00	1.50	284.70
Kothapeta	7.40	320.00	108.00	200.00	280.00	1.10	51.50
Shankarapuram	8.00	352.00	132.00	240.00	580.00	1.60	53.40
Chotapalli	8.00	420.00	140.00	380.00	1180.00	1.00	68.00
Dorasanipalli	7.80	320.00	124.00	680.00	1040.00	1.50	47.60
Ramapuram	8.00	308.00	104.00	400.00	420.00	1.00	49.50
Gopavaram	7.90	300.00	104.00	420.00	580.00	1.60	47.60
Bagath Singh Colony	8.10	400.00	160.00	320.00	320.00	1.10	58.30
Virat Nagar	7.90	300.00	100.00	360.00	180.00	0.90	48.60
Linga reddy Kottalu	7.90	592.00	196.00	520.00	236.00	1.50	96.20
Housing Board Colony	7.90	400.00	140.00	440.00	580.00	1.60	63.10
Rajiv Nagar	8.00	308.00	112.00	340.00	540.00	1.40	47.60
Dwaraka Nagar	7.70	600.00	200.00	440.00	240.00	1.40	97.20
Yanadi Colony	7.90	340.00	120.00	500.00	980.00	1.00	53.40
Srinivas Nagar	7.80	432.00	132.00	220.00	420.00	1.60	72.90
Kamanur	7.80	320.00	132.00	588.00	2640.00	1.00	45.60
Radha Nagar	7.90	420.00	132.00	440.00	520.00	0.70	69.90
Nakkaladinne	8.00	420.00	140.00	700.00	632.00	1.60	68.00
Jamespeta	8.10	420.00	160.00	300.00	700.00	1.10	63.10
Chinnamurujupalli	7.70	500.00	156.00	580.00	700.00	1.30	83.50
Nagaiahpalli	7.90	200.00	80.00	294.00	1884.00	0.60	29.10
Kakirenipalli	7.70	312.00	104.00	430.00	380.00	1.50	50.50

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Narasimhapuram	7.80	384.00	140.00	600.00	380.00	1.40	59.20
Upparapalli	7.60	304.00	104.00	770.00	332.00	1.70	48.60
Bojjavaripalli	7.70	320.00	104.00	510.00	996.00	0.80	52.40
Bankachinnaipalli	8.00	260.00	92.00	716.00	1480.00	1.00	40.80
Thallamapuram	8.00	592.00	196.00	1080.00	1160.00	0.90	96.20
Sagiliguddupalli	7.80	340.00	120.00	470.00	324.00	1.50	53.40
Mallavaripalli	8.10	840.00	380.00	592.00	300.00	1.00	111.70
Seethampalli	8.10	592.00	192.00	400.00	440.00	1.30	97.20
Kalluru	7.70	424.00	156.00	460.00	260.00	0.80	65.10
Nagganuripalli	7.50	252.00	100.00	580.00	600.00	1.30	36.90
Kothapalli	7.80	908.00	320.00	420.00	340.00	1.60	142.80
Lingapuram	8.20	392.00	132.00	220.00	1400.00	1.30	63.10
Kadarnad	8.20	320.00	108.00	420.00	620.00	1.10	51.50
Menapuram	7.80	312.00	104.00	650.00	1200.00	1.60	50.50
Kanapalli	7.80	700.00	480.00	400.00	692.00	1.10	53.40
Vivekananda nagar	8.10	696.00	220.00	436.00	432.00	1.60	115.60
Easwar Reddy nagar	7.80	220.00	88.00	360.00	2540.00	2.50	32.00
Penna Nagar	8.00	260.00	92.00	480.00	408.00	1.30	40.80
Somulavaripalli	7.90	284.00	80.00	310.00	5000.00	1.10	49.50
Settipalli	7.60	340.00	120.00	306.00	392.00	0.70	53.40
Bollavaram	7.50	232.00	76.00	320.00	532.00	0.80	37.90

Table 2: Summary statistic of hydrochemical data before log transformation

Variable	Min	Max	Mean	Median	SD	Skewness	Kurtosis
pH	7.4	8.2	7.8674	7.9	0.18609	-0.41294	2.8482
Total Hardness	200	1692	433.67	340	253.35	3.1767	15.446
Calcium	76	520	155.81	132	96.822	2.5121	8.9213
Magnesium	29.1	284.7	67.472	53.4	41.504	3.6024	18.724
Chloride	180	5000	866.98	580	892.86	2.8114	12.209
Fluoride	0.6	2.5	1.2558	1.3	0.38273	0.65584	4.412
Alkalinity	200	1080	455.86	430	171.07	1.2363	5.4526

Table 3: Summary statistic of hydrochemical data after log transformation

Variable	Min	Max	Mean	Median	SD	Skewness	Kutosis
pH	2.0015	2.1041	2.0625	2.0669	0.023782	-0.47249	2.92.5
Total Hardness	5.2983	7.4337	5.9703	5.8289	0.41637	1.2732	5.1121
Calcium	4.3307	6.2538	4.9309	4.8828	0.44305	1.4094	4.7874
Magnesium	3.3707	5.6514	4.1064	3.9778	0.41889	1.3431	5.8561
Chloride	5.193	8.5172	6.4469	6.363	0.74515	0.7656	3.1036
Fluoride	-0.51083	0.9162	0.18658	0.26236	0.29434	-0.26774	2.8575
Alkalinity	5.2983	6.9847	6.0587	6.0638	0.35933	0.0569	2.9849

Table 4: Correlation matrix

Correlation matrix (Pearson (n)):							
Variables	PH	TH	Ca	ALKALINITY	CL	F	Mg
PH	1	0.167	0.150	-0.024	0.088	0.010	0.162
TH	0.167	1	0.905	-0.014	0.029	0.119	0.970
Ca	0.150	0.905	1	-0.003	-0.027	0.058	0.775
ALKALINITY	-0.024	-0.014	-0.003	1	-0.051	0.026	-0.019
CL	0.088	0.029	-0.027	-0.051	1	0.025	0.058
F	0.010	0.119	0.058	0.026	0.025	1	0.144
Mg	0.162	0.970	0.775	-0.019	0.058	0.144	1
Values in bold are different from 0 with a significance level alpha=0.05							

Table 5: Factor analysis

Maximum change in communality at each iteration:	
Iteration	Maximum change
40	0.0005
41	0.0005
42	0.0005
43	0.0004
44	0.0004
45	0.0004
46	0.0004
47	0.0004
48	
49	

Table 6: Eigen values

	F1	F2	F3	F4	F5
Eigenvalue	2.802	0.287	0.174	0.030	0.002
Variability (%)	40.029	4.097	2.487	0.424	0.034
Cumulative %	40.029	44.127	46.614	47.038	47.071

Table 7: Best interpolation method based on RMSE

GWQI	IDW	RBF	Kriging
pH	0.1953	0.1942	0.1856
TH	284.5	271.6	308.9
Ca	105.7	102.3	108.2
Mg	46.88	44.61	51.45
Cl	964	911.9	985.7
F	0.3839	0.3813	0.3851
Alkalinity	173.6	172.5	173.8

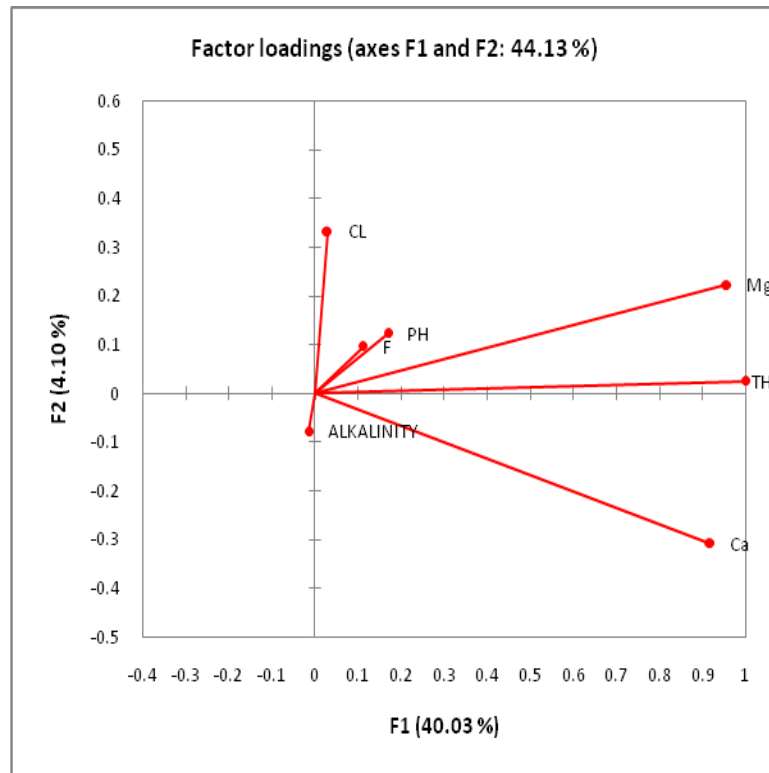


Figure 2: Factor loadings

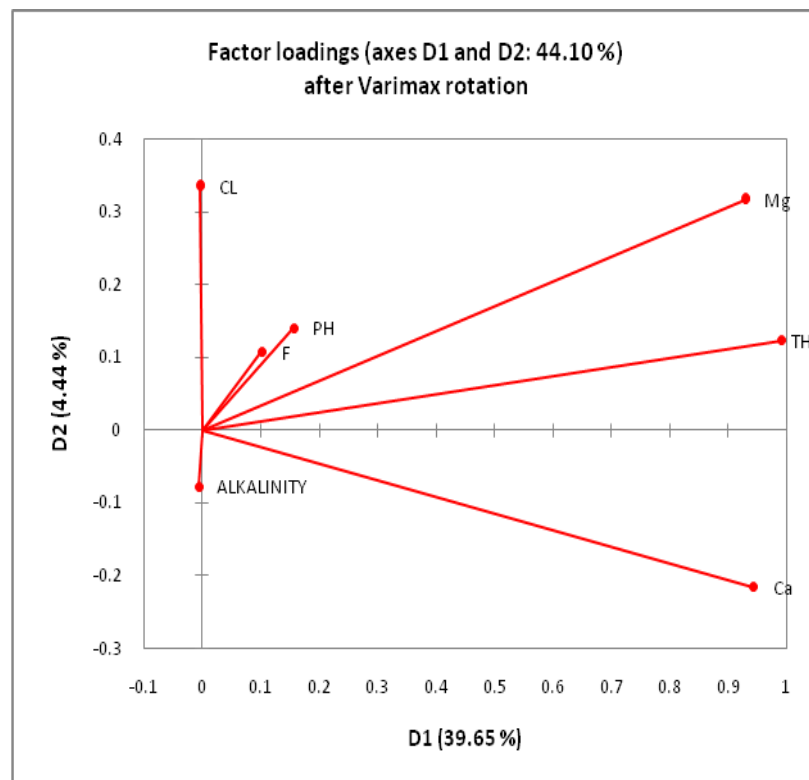


Figure 3: Factor loadings after varimax rotation

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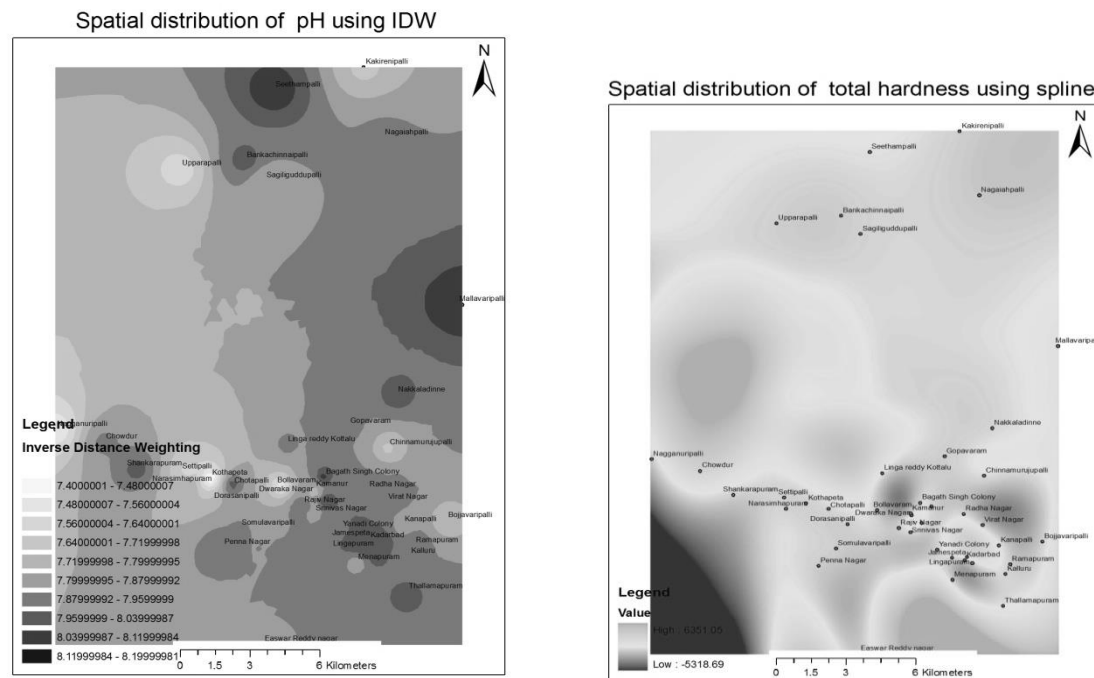


Figure 4: Spatial distributions of pH and total hardness

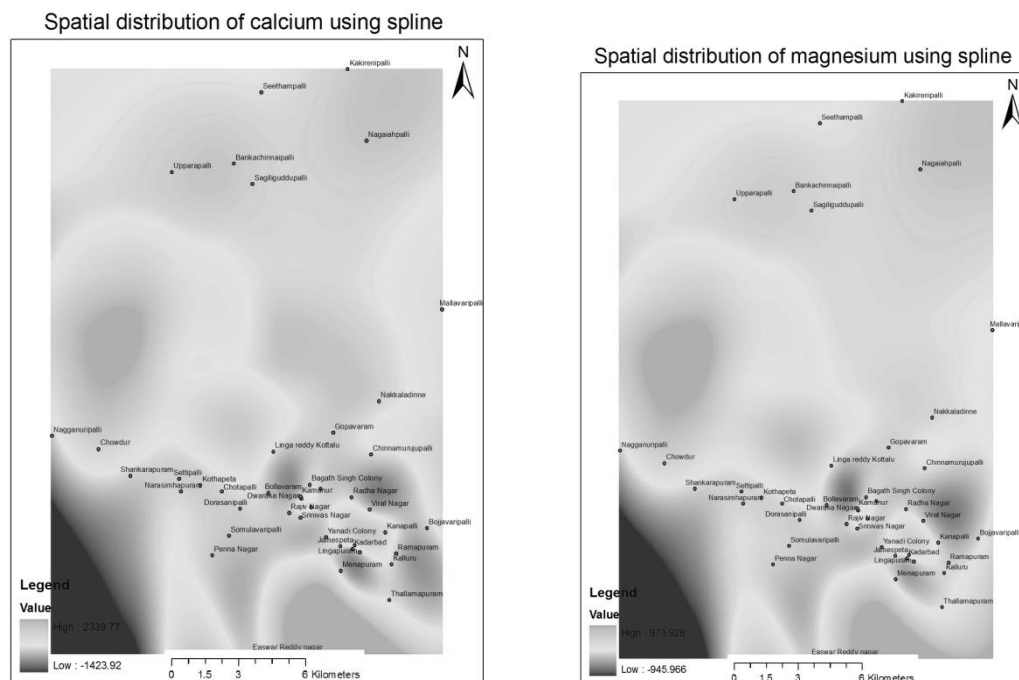


Figure 5: Spatial distributions of Calcium and Magnesium

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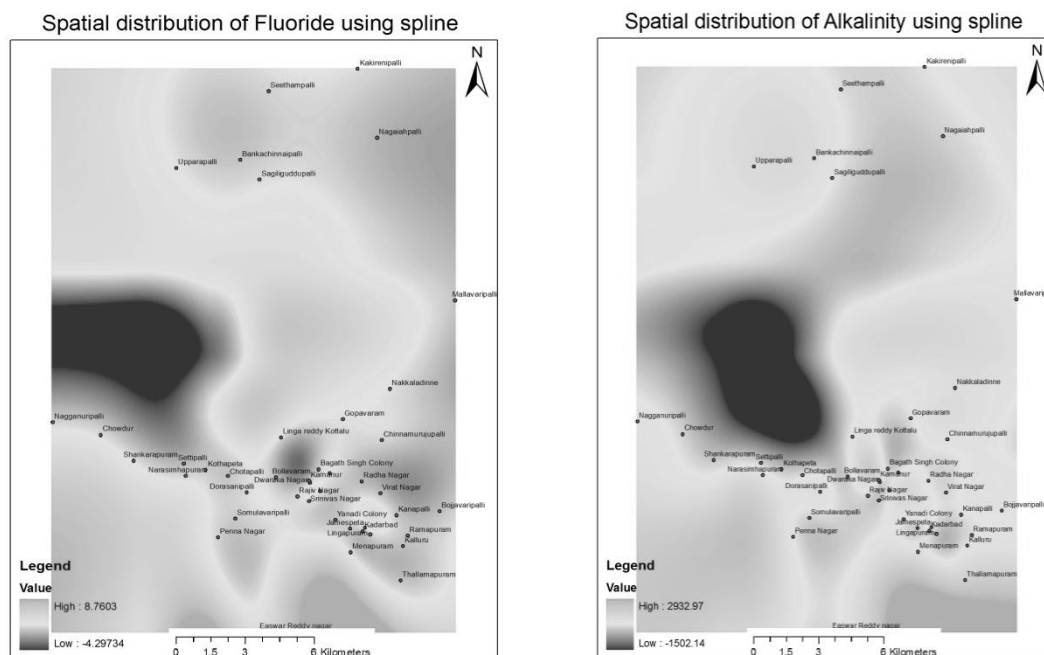


Figure 6: Spatial distributions of Fluoride and Alkalinity

5. Conclusion

The statistical treatment was applied to the hydro geochemical data via principal component analysis and factor analysis. Three interpolation techniques were critically analyzed and the best fit was selected after evaluating RMSE values. As per the magnitude of variations and prominence offered by the data, it is observed that except pH, all the other variables such as Total hardness, Calcium, Magnesium, Chloride, Fluoride and alkalinity accepts radial basis function method of interpolation and pH sticks on to the Inverse Distance Weighted method of interpolation. GIS maps were prepared for groundwater quality parameters.

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